ASSIGNMENT 3

Date: 17/09/2026

1. AIM AND OBJECTIVES

1. To study and analyze Local Oxidation of Silicon (LOCOS) and Shallow Trench Isolation (STI) techniques using Semulator3D.
2. To study the effect of isolation thickness on device performance, determine the optimal active area, and measure the optimal gate area and isolation oxide thickness (Assignment 3.1.a).
3. To study the relationship between planned gate pitch and gate length, determine the smallest achievable gate pitch, and report the corresponding gate length (Assignment 3.1.b).

2. TOOL USED

Coventor Semulator3D (3D Process Modeling and Virtual Fabrication Software).

3. THEORY OF LOCOS

Isolation is an important part of semiconductor device fabrication. It electrically separates adjacent devices, minimizes leakage current, provides structural support, and ensures reliable device operation.

This assignment investigates the LOCOS and STI isolation techniques and analyzes the impact of oxide thicknesses and gate pitch on LOCOS performance.

The study involves identifying the isolation oxide, gate oxide, gate length, gate pitch, and bird's beak. The achieved gate length is compared with the planned gate length using simulation results and graphs.

Local Oxidation of Silicon (LOCOS):

LOCOS is a traditional isolation technique in which a thick oxide is thermally grown in selected regions of a silicon wafer to electrically isolate adjacent devices.

A silicon nitride layer is used as an oxidation mask. A thin pad oxide layer of approximately 15 nm is placed between the silicon substrate and silicon nitride layer to reduce mechanical stress.

The oxidation process produces a tapered oxide region under the edge of the nitride mask, known as the **bird's beak**.

4. LOCOS PROCESS RECIPE

The LOCOS process is implemented in Coventor Semulator3D using the Process Editor. The following table represents the sequence of fabrication steps used in the simulation.

6. THICKNESS SPECIFICATION AND DEVICE DIMENSIONS

Gate oxide is the pad oxide deposited before the silicon nitride layer above the active area.

Isolation oxide is the thick thermally grown oxide that provides electrical insulation between adjacent devices.

The following table compares the desired nominal specifications with the measured dimensions obtained from Semulator3D.

7. QUESTION 1(A): OPTIMAL ACTIVE AREA

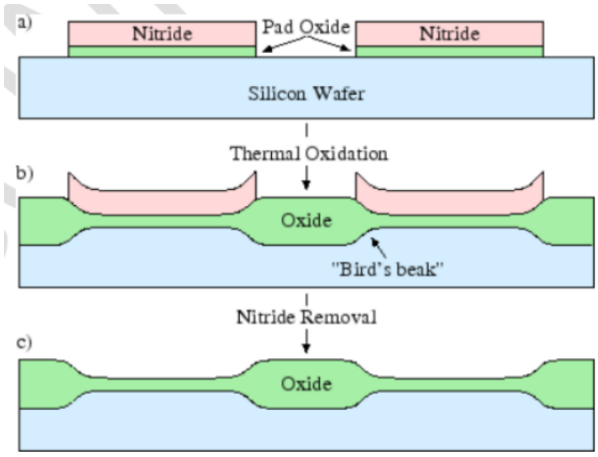
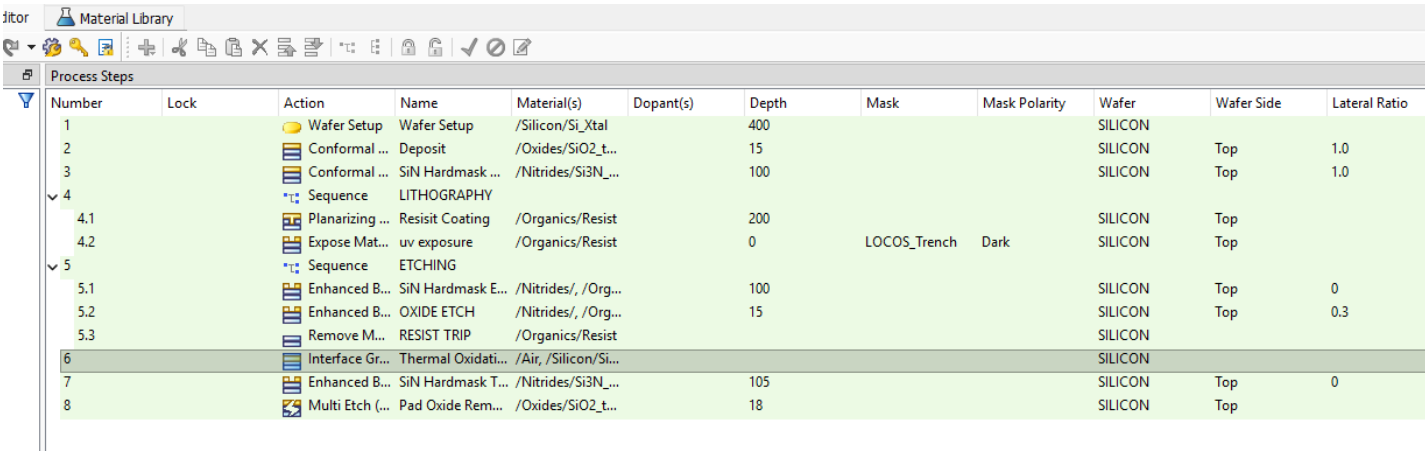


Figure 1: LOCOS Process Flow



Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		400			SILICON		
2		Conformal ...	Deposit	/Oxides/SiO2_t...		15			SILICON	Top	1.0
3		Conformal ...	SiN Hardmask ...	/Nitrides/Si3N_...		100			SILICON	Top	1.0
4		Sequence	LITHOGRAPHY								
4.1		Planarizing ...	Resisit Coating	/Organics/Resist		200			SILICON	Top	
4.2		Expose Mat...	uv exposure	/Organics/Resist		0	LOCOS_Trench	Dark	SILICON	Top	
5		Sequence	ETCHING								
5.1		Enhanced B...	SiN Hardmask E...	/Nitrides/, /Org...		100			SILICON	Top	0
5.2		Enhanced B...	OXIDE ETCH	/Nitrides/, /Org...		15			SILICON	Top	0.3
5.3		Remove M...	RESIST TRIP	/Organics/Resist					SILICON		
6		Interface Gr...	Thermal Oxidati...	/Air, /Silicon/Si...					SILICON		
7		Enhanced B...	SiN Hardmask T...	/Nitrides/Si3N_...		105			SILICON	Top	0
8		Multi Etch (...	Pad Oxide Rem...	/Oxides/SiO2_t...		18			SILICON	Top	

Figure 2: Steps involved in LOCOS Process (Process Editor)

Table 1: Steps involved in LOCOS Process (Process Editor Recipe)

Step	Action	Step Name	Material(s)	Depth (nm)	Mask	Polarity
1	Wafer Setup	Wafer Setup	/Silicon/Si_Xtal	400	-	-
2	Conformal Deposit	Pad oxide deposition	/Oxides/Silicon_Oxide_Thermal	15	-	-
3	Conformal Deposit	Silicon nitride hard mask deposition	/Nitrides/Silicon_Nitride_LPCVD	100	-	-
4	Sequence	Lithography	-	-	-	-
4.1	Planarizing Deposit	Deposit	/Organics/Resist	200	-	-
4.2	Expose Material	UV exposure	/Organics/Resist	0	LOCOS_Trench	Dark
5	Sequence	Etch	-	-	-	-
5.1	Enhanced Basic Etch	SiN hardmask etch	/Nitrides/, /Organics/, /Oxides/	100	-	-
5.2	Enhanced Basic Etch	Pad oxide etch	/Nitrides/, /Organics/, /Oxides/	15	-	-
5.3	Remove Materials	Resist removal	/Organics/Resist	-	-	-
6	Interface Growth	Thermal oxidation	/Air, /Silicon/Si_Xtal	-	-	-
7	Enhanced Basic Etch	Silicon nitride hardmask strip	/Nitrides/Silicon_Nitride_LPCVD	105	-	-
8	Multi Etch	Multi Etch	/Oxides/Silicon_Oxide_Thermal	18	-	-

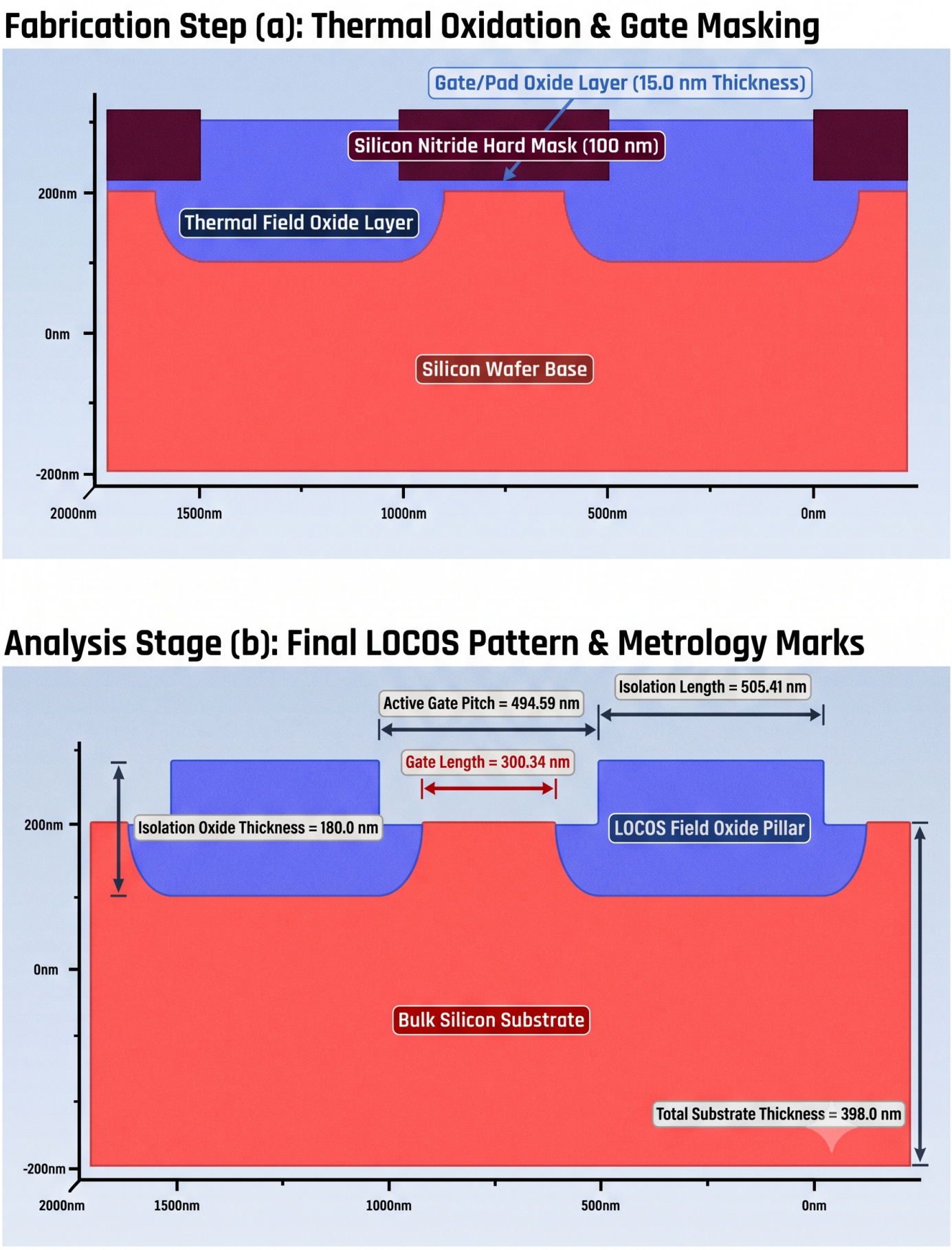


Figure 3: Simulated Gate and Isolation Oxide Dimensions

Table 2: Desired and Measured Device Dimensions

S.No	Parameter	Desired Value (nm)	Measured Value (nm)
01	Wafer	400	398.0
02	Gate Oxide (Pad Oxide)	15	15.0
03	Isolation Oxide	100	180.0
04	Gate Length	300	300.34
05	Gate Pitch	350	494.59
06	Isolation Length	100	505.41

LOCOS utilizes the difference in oxidation rates between silicon and silicon nitride. The silicon nitride layer acts as an oxidation mask, while the thin pad oxide layer reduces mechanical stress at the silicon-nitride interface.

During thermal oxidation, the field oxide grows around the active regions and provides electrical isolation between adjacent devices.

However, oxidants can diffuse laterally beneath the edge of the nitride mask through the pad oxide. This produces a tapered oxide region called the **bird's beak**.

Limitations of LOCOS:

1. **Bird's Beak Formation:**

   Lateral oxidant diffusion beneath the nitride mask causes oxide growth under the mask edge. This reduces the effective active area.

2. **Scalability Problem:**

   As device dimensions decrease, lateral encroachment becomes a significant fraction of the active area. This reduces the effective gate length below the planned mask dimension.

Possible Mitigation:

Higher oxidation temperatures can increase the vertical oxidation rate relative to lateral diffusion. Another alternative is Shallow Trench Isolation (STI), where vertical trenches are etched into silicon, filled with deposited oxide, and planarized using CMP.

The purpose of this assignment is to determine the optimal active area where the field oxide provides sufficient dielectric isolation without excessive reduction in active gate length.

Gate length depends on planned gate pitch, isolation oxide thickness, and isolation length.

8. ASSIGNMENT 3.1(A): EFFECT OF ISOLATION THICKNESS

The effect of isolation oxide thickness on active gate length was studied by performing simulations for different desired isolation oxide thicknesses ranging from 100 nm to 320 nm.

The simulation results are presented in Table 2.

Analysis:

Table 3: Effect of Isolation Oxide Thickness on Device Dimensions

S.No	Desired Isolation Oxide Thickness (nm)	Measured Isolation Depth (nm)	Measured Gate Length (nm)	Measured Isolation Length (nm)
1	100	80.0	400.84	505.51
2	120	100.0	380.74	505.51
3	150	130.0	350.72	505.51
4	180	160.0	320.72	505.51
5	200	180.0	300.72	505.51
6	300	280.0	200.72	505.51
7	320	300.0	180.72	505.51

1. Simulations were performed for different isolation oxide thicknesses between 100 nm and 320 nm.
2. As the measured isolation depth increases from 80.0 nm to 300.0 nm, the measured active gate length decreases from 400.84 nm to 180.72 nm.
3. This reduction occurs because of lateral oxidant diffusion beneath the silicon nitride hardmask, which reduces the active region.
4. The optimal isolation oxide thickness is reported as 180.0 nm, corresponding to an active gate length of 300.34 nm and an isolation length of 505.41 nm.

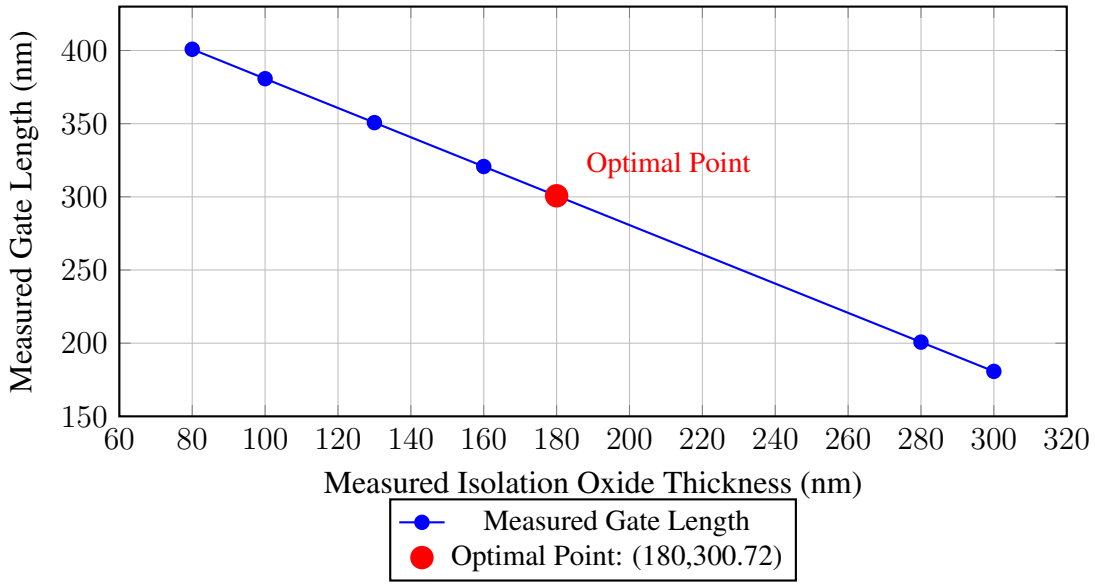


Figure 4: Effect of Isolation Oxide Thickness on Measured Gate Length

9. QUESTION 1(B) / ASSIGNMENT 3.1(B): PLANNED GATE PITCH VS. GATE LENGTH

The relationship between planned gate pitch and achieved gate length was studied by varying the planned gate pitch from 350 nm to 150 nm.

Isolation oxide thickness entered = 100 nm.

The simulation results are presented in the following table.

Table 4: Relationship Between Planned Gate Pitch and Measured Gate Length

S.No	Planned Gate Pitch (nm)	Measured Isolation Depth (nm)	Measured Gate Length (nm)	Measured Isolation Length (nm)
1	350	80.0	250.84	505.51
2	300	80.0	200.84	505.51
3	250	80.0	150.84	505.51
4	200	80.0	100.84	505.51
5	150	80.0	50.84	505.51

Analysis:

1. **Scaling Relationship:**

   Across planned gate pitches ranging from 350 nm to 150 nm, the achieved gate length follows a linear relationship.

   $$\text{Planned Gate Pitch} = \text{Gate Length} + 93.65 \text{ nm}$$

   The constant offset of approximately 93.65 nm represents the process margin between active gates.

2. **Smallest Achievable Gate Pitch:**

   The smallest achievable gate pitch is 144.49 nm, corresponding to a gate length of 50.84 nm.

3. **Physical Scaling Limit:**

   Below a gate length of approximately 50 nm, opposing bird's beak regions merge, resulting in complete pinch-off and loss of active silicon.

10. OBSERVATIONS AND DISCUSSION

1. **Stress Relief and Lateral Diffusion Trade-Off:**

   The 15 nm pad oxide layer reduces mechanical stress at the silicon-nitride interface. However, it also provides a path for lateral oxidant diffusion during high-temperature field oxidation.

2. **Isolation Oxide and Active Dimension Trade-Off:**

   Increasing isolation oxide thickness improves dielectric isolation and breakdown voltage. However, it also reduces the effective active gate length from 400.84 nm to 180.72 nm.

   The optimal baseline is reported at an isolation depth of 180.0 nm with a gate length of 300.34 nm.

3. **Pitch Scaling Boundary:**

   Gate pitch scales linearly down to 144.49 nm at a gate length of 50.84 nm.

   Below this physical limit, lateral oxidation can destroy the active device region.

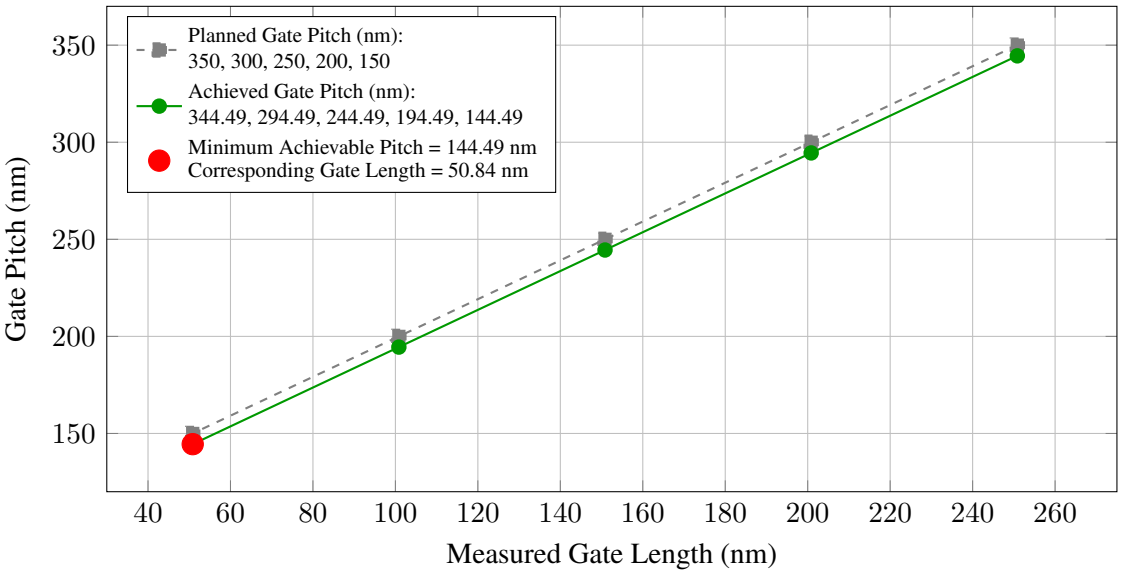


Figure 5: Planned and Achieved Gate Pitch Versus Gate Length

11. CONCLUSION

The LOCOS fabrication sequence was successfully modeled and analyzed using Coventor Semulator3D. The virtual fabrication process demonstrated selective thermal oxidation and lateral oxide encroachment beneath the silicon nitride mask.

The simulation results show that increasing the field oxide thickness reduces the active gate length because of lateral oxidation. The optimal operating point is reported as 180.0 nm isolation oxide thickness and 300.34 nm gate length.

The gate pitch analysis identifies the smallest achievable gate pitch as 144.49 nm, corresponding to a gate length of 50.84 nm.

This experiment demonstrates the scaling limitations of the LOCOS isolation technique and explains why advanced semiconductor technology nodes adopt Shallow Trench Isolation (STI) for dense integrated circuits.

Signature: _____

EXPERIMENT NO. 8

Date: 09/10/2026

1. Aim

To demonstrate the physical vapour deposition (PVD) of a tin dioxide (SnO_2) target onto an ITO-coated glass substrate using a sputtering system.

2. Apparatus Required

The following apparatus and materials are required for the sputtering process:

1. **Sputtering System:** Consists of RF and DC power sources, a rotary pump, and a turbo-molecular pump.
2. **Chiller:** Used for cooling the sputtering system during operation.
3. **Venting Pump:** Used for the venting process.
4. **Argon Gas:** Used as the sputtering gas with a flow rate of 5 sccm for foramtion of plasma.
5. **Substrate:** ITO-coated glass.
6. **Target:** Tin dioxide (SnO_2).

3. Theory

Physical Vapour Deposition (PVD):

Physical Vapour Deposition (PVD) is a thin-film deposition technique in which a material is physically transferred from a source (target) to a substrate under controlled vacuum conditions. In this experiment, tin dioxide (SnO_2) is used as the target material, and ITO-coated glass is used as the substrate. The deposition is carried out using a sputtering system.

Principle of Sputtering:

Sputtering is a PVD technique in which energetic ions bombard the surface of a solid target and eject atoms or particles from it. These ejected particles travel through the vacuum chamber and deposit onto the substrate, forming a thin film.

Argon (Ar) gas is introduced into the vacuum chamber at a controlled flow rate of 5 sccm. A voltage is applied between the electrodes using an RF or DC power source, which ionizes the argon gas and produces a plasma. The plasma consists of positively charged argon ions and free electrons.

The positively charged argon ions are accelerated towards the negatively biased target. When these ions strike the SnO_2 target, they transfer their kinetic energy to the target atoms, causing them to be ejected from its surface. These sputtered particles then travel towards the ITO-coated glass substrate and gradually form a thin SnO_2 film.

Role of Different Components:

1. **Target (SnO_2):** It is the source material from which atoms are ejected during the sputtering process.
2. **Substrate (ITO-coated Glass):** It provides the surface on which the sputtered SnO_2 particles are deposited to form a thin film.

3. **Argon Gas:** Argon is an inert gas used to generate plasma. Its ions bombard the target and cause sputtering.
4. **RF/DC Power Source:** It supplies electrical power to generate and sustain plasma inside the chamber.
5. **Rotary and Turbomolecular Pumps:** These pumps create the required vacuum inside the chamber. The rotary pump provides initial rough vacuum, while the turbomolecular pump helps achieve a higher vacuum.
6. **Chiller:** It maintains the required cooling of the sputtering system during operation.

4. Procedure

The following steps are performed to deposit a thin film of SnO_2 on an ITO-coated glass substrate using the RF sputtering technique.

1. Load the ITO-coated glass substrate into the sputtering chamber. If the chamber is under vacuum, first vent the chamber by opening the vent valve. After venting is completed, close the vent valve.
2. Switch ON the Pirani gauge to monitor the vacuum pressure inside the chamber.
3. Turn ON the rotary pump by setting its selector switch to the ON position.
4. Open the backing valve to establish the required backing pressure.
5. After achieving the required backing pressure, switch ON the turbomolecular pump and follow the pumping sequence specified in the equipment manual.
6. Open the roughing valve to evacuate the chamber. Once the required roughing pressure is achieved, operate the valves according to the prescribed vacuum sequence.
7. Switch ON the high-vacuum gauge and open the high-vacuum valve. Allow the turbomolecular pump to evacuate the chamber until the required high-vacuum pressure is achieved.
8. Ensure that the SnO_2 target is properly mounted in the sputtering system.
9. Open the argon gas valve and switch ON the required mass flow controller. Set the argon gas flow rate to 5 sccm.
10. Switch ON the RF power supply and set the required RF power according to the experimental setup.
11. Allow the argon gas to ionize and generate plasma inside the chamber. The positively charged argon ions bombard the SnO_2 target and eject its atoms.
12. Operate the shutter to expose the ITO-coated glass substrate to the sputtered particles. The particles gradually deposit on the substrate and form a thin SnO_2 film.
13. Monitor the deposition process using the thickness monitor. Once the required coating thickness is achieved, close the shutter.
14. Switch OFF the RF power supply and close the argon gas valve.
15. Allow the system to cool if required. Follow the prescribed venting procedure to bring the chamber to atmospheric pressure before removing the coated substrate.

5. Equipment Images



Sputtering System



Venting Pump



Chiller



Control Panel

6. Parameters

The following parameters were used during the SnO_2 thin-film deposition process:

S.No.	Parameter	Value
1	Thickness of Film	90 nm
2	Power of RF Supply	90 W
3	Flow Rate of Argon Gas	5 sccm

Table 5: Parameters Used for SnO_2 Deposition

7. Observations

During the experiment, plasma formation was observed inside the sputtering chamber after introducing argon gas and applying RF power. The plasma generated energetic argon ions, which bombarded the SnO_2 target and caused the ejection of target particles. These particles travelled towards the ITO-coated glass substrate and gradually formed a thin film.

The sputtering process was carried out under controlled vacuum conditions with an argon gas flow rate of 5 sccm and an RF power of 90 W. The experiment demonstrated the role of plasma generation, target bombardment, and controlled deposition in the formation of thin films using the RF sputtering technique.

8. Conclusion

The experiment provided practical understanding of the RF sputtering technique, which is an important method for depositing thin films in semiconductor and electronic device fabrication. It demonstrated how process parameters such as RF power, gas flow rate, and deposition conditions influence the thin-film formation process. The experiment also helped in understanding the importance of vacuum systems, plasma generation, and target-substrate interaction in physical vapour deposition. Overall, the experiment enhanced practical knowledge of thin-film fabrication and the working principle of a sputtering system.

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